

Rachel Peterson

rachel.peterson.5683@gmail.com

[LinkedIn](#) | [Github](#)

EXPERIENCE

Software Engineer / Associate Scientist II

May 2026 - Present

Software Engineer / Associate Scientist I

November 2024 - May 2026

Cooperative Institute for Research in Environmental Sciences

- Architecting and developing the AWS cloud migration of NOAA National Centers for Environmental Information's ~30TB marine trackline geophysical data archive and associated QA/QC pipelines, leveraging Python, GitLab, S3, CloudFormation, Step Functions, Lambda and other technologies to improve scalability and reliability
- Maintaining and improving full-stack Java, React, and Vue-based cloud-native and on-premises applications used in the management of marine geophysical data
- Implementing existing best practice metadata standards and developing cloud-native data format translations for marine geophysical datasets
- Organizing and facilitating regular cross-team technical working groups to enhance organizational communication and knowledge-sharing
- Consulting on cloud migration technical strategy for the ocean video data holdings at NOAA NCEI
- Connecting scientific dataset expertise with enterprise-level technical knowledge to effectively communicate and implement user needs in collaboration across many development teams in an Agile context

Production Support Analyst

Connect For Health Colorado

January 2023 - October 2024

- Served as a product subject matter expert and liaison between support and development teams, ensuring compliance with state and federal health policy
- Identified and resolved thousands of accounts impacted by application bugs, including careful consultation with legal personnel to resolve system issues with health policy compliance implications
- Recognized patterns across large numbers of customer account issues and connected application expertise with technical understanding, ultimately resulting in the remediation of thousands of customer account issues

Configuration Analyst

Shelter Insurance Companies

October 2021 - January 2023

- Supported product upgrades across multiple code releases by coordinating testing across product areas and facilitating and testing code deployments
- Developed, executed and maintained manual test scripts, reducing post-release defects
- Analyzed test results and created reports for efficient communication of product defects to vendor

Teaching Assistant

Oregon State University

June 2021 - December 2021

- Supported over 100 undergraduates in Discrete Structures through office hours and online forums
- Simplified complex theoretical concepts into accessible teaching, boosting student

comprehension and engagement

SEVIS Compliance Assistant

University of Missouri - Columbia

November 2019 - January 2021

- Maintained updated records for 1000+ international students and organized a student records database of 30,000+ documents to ensure university compliance with federal regulations

Sustainability Office Assistant

University of Missouri - Columbia

May 2017 - August 2018

- Assisted in the implementation of campus-wide sustainability programming
- Monitored the sustainability performance of the MU system and implemented initiatives in areas where improvement was needed

Undergraduate Research Fellow

University of Missouri - Columbia

August 2015 - March 2017

- Recruited and interviewed participants and completed data entry for multiple studies in the Physical Therapy department under the direction of a faculty mentor
- Presented independent secondary analysis at local and national research conferences

VOLUNTEER EXPERIENCE

Board Trustee, Climate Action Ministry Coordinator

Unitarian Universalist Church of Boulder

October 2022 - Present

- Serving on the Board of Trustees – providing strategic guidance, organizational governance, and oversight of a budget for a community of a few hundred members
- Coordinating climate action initiatives, including volunteer recruitment, programming, and facilitation for congregational engagement campaigns

Executive Board Project Coordinator

Environmental Leadership Ambassadors

August 2015 - May 2017

- Presented to various university student organizations on topics involving sustainability and environmentalism
- Coordinated ambassador project leaders and their respective projects promoting campus sustainability efforts

Project Lead

Sustain Mizzou

August 2015 - December 2018

- Coordinated a university-wide electronic waste drive
- Participated in various local sustainability-oriented volunteer projects

SKILLS

Python, AWS (Cloudformation, Step Functions, S3, EC2, Lambda, IAM), Java, SQL, Javascript (React), Bash, Neo4J, Gitlab, Github, Agile, Spanish Professional Working Proficiency

EDUCATION

University of Colorado, Boulder | August 2026 - Present

Master of Science in Information Science

Oregon State University | March 2021 - June 2024

Bachelor of Science in Computer Science | GPA: 4.0

University of Missouri, Columbia | August 2015 - December 2018

Bachelor of Arts in Romance Languages - Spanish | GPA: 3.99

PUBLICATIONS

B. Bock, D. Van Bramer, **R. Peterson**. “*Environmental Footprint Reporting as a Performance Metric in Environmental Data Science*,” *Environmental Data Science* (under review, submitted September 2026).

PRESENTATIONS

R. Peterson, B. Meyer. “*Transitioning Data to the Cloud*,” GEBCO Nippon Scholars Visit, Boulder, CO (May 2025).

R. Peterson. “*Transitioning Data to the Cloud*,” GEBCO Nippon Scholars Visit, Boulder, CO (July 2026).

R. Peterson. “*Modernizing NOAA’s Legacy Marine Trackline Geophysical Data: Translating Bathymetry, Gravity, and Magnetic Data into Cloud-native GeoParquet Formats*”, NOAA Enterprise Data Management Working Group (EDMW) (abstract submitted, 2026).

POSTERS

R. Peterson, B. Meyer, J. Schweizer, T. Mitroi, M. Royer, R. Klucik. “*Migrating Marine Trackline Geophysical Data to the Cloud: A Paradigm Shift for Data Management*,” American Geophysical Union, New Orleans, LA (December 2025).

R. Peterson. “*Modernizing NOAA’s Legacy Marine Trackline Geophysical Data: Translating Bathymetry, Gravity, and Magnetic Data into Cloud-native GeoParquet Formats*”, American Geophysical Union, virtual poster (abstract submitted, 2026).